COM668 Computing Project
AT1 Project Proposal Form

****** Important note, you must agree your proposal with your mentor before submission ******

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Mentor name:	**Ayobami Godwin Akindele**	PSG:	**PSG-Q12**

Project Title: Medication Prescribing Data Warehouse with Analytics Dashboard

Declaration: *"I declare that this is all my own work. Any material I have referred to has been accurately referenced and any contribution of Artificial Intelligence technology has been fully acknowledged. I understand the importance of academic integrity and have read and understood the University's General Regulation: Student Academic Integrity and the Academic Misconduct Procedure. I understand that I must not upload my work before, during or after submission to any unapproved plagiarism detectors or answer sharing platforms, or equivalent, and that only University-approved platforms should be used."*

Important note, by submitting this work you are agreeing to the above declaration.

Aim (50 words):

To design and implement a dimensional star schema data warehouse using PostgreSQL and a Python-based Extract, Transform, Load (ETL) pipeline that processes anonymised NHS prescribing data and delivers analytical insights through optimised SQL queries and an interactive Streamlit dashboard.

Problem description (100 words):

To design and implement a dimensional star schema data warehouse using PostgreSQL and a Python-based Extract, Transform, Load (ETL) pipeline that processes anonymised NHS prescribing data and delivers analytical insights through optimised SQL queries and an interactive Streamlit dashboard.

Intended outcomes (100 words):

The project will deliver:

• A dimensional star schema data warehouse including:

- *fact_prescribing*
- *dim_medication*
- *dim_practice*
- *dim_date*

• A structured Python ETL pipeline with:

- *Extraction from NHS open datasets*
- *Data cleaning (null handling, standardisation, duplicate removal)*
- *Validation rules (range checks, foreign key integrity)*
- *Transformation into dimensional format*

- *Controlled loading into PostgreSQL*

• Indexed and performance-optimised database

• Example analytical SQL queries

• A Streamlit dashboard enabling interactive exploration of prescribing trends

• Testing and verification documentation

Challenge suitability (50 words):
The project aligns with Computing learning outcomes and British Computer Society expectations by demonstrating database design, dimensional modelling, Python ETL engineering, query optimisation, validation strategy, and interactive analytics delivery. It integrates software development, data engineering, and system testing practices suitable for a final-year Computing project.

Copyright/IPR/Commercial Sensitivity (50 words):
All datasets used will be publicly available NHS prescribing datasets containing no identifiable patient information. The data is aggregated at practice level and compliant with open data licensing. All database designs, ETL scripts, and dashboard components will be independently developed. No commercial or confidentiality restrictions apply

Resources required (50 words):

Hardware: Personal laptop or university laboratory computer.

Software: PostgreSQL, Python (pandas, psycopg2), Streamlit.

Data: NHS Open Prescribing dataset.

All resources are freely available. No financial or ethical restrictions are anticipated due to use of anonymised public data.

References (use Harvard style):
NHS Business Services Authority (2023) NHS Prescribing Data. Available at: https://www.nhsbsa.nhs.uk (Accessed: [insert date]).